

Surface plasmons in cylindrical systems

In the below, $\epsilon_m = n_m^2$ and $\epsilon_d = n_d^2$ are the permittivities of the cylinder interior and exterior, respectively.

The Laplacian in cylindrical systems is given by:

$$\frac{1}{\rho} \partial_\rho (\rho \partial_\rho \Psi(\rho, \phi, z)) + \frac{1}{\rho^2} \partial_\phi^2 \Psi(\rho, \phi, z) + \partial_z^2 \Psi(\rho, \phi, z) = 0$$

We take for the scalar potential, $\Psi(\mathbf{r}) = f(\rho) e^{im\phi} e^{ikz}$. For $k = 0$, we have:

$$\rho \partial_\rho (\rho \partial_\rho f(\rho)) = m^2 f(\rho) \rightarrow \Psi(\rho, \phi, z) = \left[\frac{a_m}{\rho^m} + b_m \rho^m \right] e^{im\phi}$$

Therefore, for $\rho < a$, we take the solution to be $b_m \rho^m$ and outside the sphere we take the solution to be $a_m \rho^{-m}$. We require the potential to match at the boundary (equivalently, we require the tangential component of the electric field to be continuous across the boundary), which gives us:

$$a_m a^{-m} = b_m a^m \rightarrow a_m = b_m a^{2m}$$

If we have no finite surface impedance, the second boundary condition will be given by:

$$-\epsilon_d a_m a^{-m-1} = \epsilon_m b_m a^{m-1} \rightarrow \epsilon_m = -\epsilon_d$$

Therefore, for a Drude model, we have $\omega_{sp}^2 = \omega_p^2/2$, where ω_p is the bulk plasmon frequency and ω_{sp} is the surface plasmon frequency.

Consistency with the retarded result

The retarded scattering function tells us that the plasmon will appear at the frequency which satisfies (see <https://hackmd.io/@aligho/B1Gs5Cz2el> (<https://hackmd.io/@aligho/B1Gs5Cz2el>)):

$$n_d J'_m(n_m ka) H_m^{(1)}(n_d ka) - n_m J_m(n_m ka) H_m^{(1)}(n_d ka)' = 0$$

In the quasistatic limit, only the Bessel function of the second kind contributes in the Hankel function (since the Bessel function of the second kind will diverge at the origin), and since $J_m \approx z^m$ and $Y_m \approx z^{-m}$ (constants of proportionality do not matter since we have zero on the RHS of the equation and each term on the LHS has exactly one J_m and one Y_m), we have:

$$n_d (n_m ka)^{m-1} (n_d ka)^{-m} + n_m (n_m ka)^m (n_d ka)^{-m-1} = 0 \rightarrow \\ n_d n_m^{-1} + n_m n_d^{-1} = 0 \rightarrow n_d^2 + n_m^2 = 0 \rightarrow \varepsilon_d + \varepsilon_m = 0,$$

which is precisely the quasistatic relation derived in the previous section.

Presence of finite surface impedance

We now consider the situation with finite surface impedance. In this case, the continuity of the potential (equivalently, the continuity of the tangential part of the electric field) still gives us:

$$a_m a^{-m} = b_m a^m \rightarrow a_m = b_m a^{2m}$$

If we have a finite surface impedance, the second boundary condition will, however, be modified and will be given by:

$$\varepsilon_d a_m a^{-m-1} + \varepsilon_m b_m a^{m-1} = \rho_s / m \varepsilon_0 \rightarrow (\varepsilon_d + \varepsilon_m) b_m a^{m-1} = \rho_s / m \varepsilon_0,$$

where $\rho_s e^{im\phi}$ is the surface charge density defined by $\tilde{\rho}(\rho, \theta, z) = \rho_s e^{im\phi} \delta(\rho - a)$, and $\tilde{\rho}$ is the volume charge density, that is determined from the conductivity, $\sigma(\omega)$, via the continuity equation:

$$\tilde{\rho}(\mathbf{r}, \omega) = -\frac{i}{\omega} \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) = \frac{i}{\omega a^2} \partial_\phi (\sigma(\omega) \partial_\phi (b_m a^m e^{im\phi} \delta(\rho - a))) = \\ \frac{i\sigma(\omega)}{\omega a^2} \partial_\phi^2 (b_m a^m e^{im\phi}) \delta(\rho - a)$$

Therefore, we get the following dispersion relation:

$$\left[\varepsilon_d + \varepsilon_m \right] a^{m-1} = -i(m^2 \sigma(\omega) / m \omega a^2 \varepsilon_0) a^m \rightarrow \varepsilon_d + \varepsilon_m = -\frac{im\sigma(\omega)}{\omega a \varepsilon_0}$$

Dispersion for graphene coated system (intraband)

In the vein of the set of notes, https://hackmd.io/@aligho/ByRI-iJ_ge (https://hackmd.io/@aligho/ByRI-iJ_ge), we see that for graphene doped to Fermi level, ε_F , the dispersion is approximately given by:

$$\varepsilon_d + \varepsilon_m = \frac{800\alpha m \varepsilon_F}{(\hbar\omega)^2 a},$$

where a is given in nanometers and energy quantities are given in eV. In deriving this result, we used the following for the graphene conductivity (taking only intraband transitions into account):

$$\sigma(\omega) \approx \frac{ie^2 \varepsilon_F}{\pi \hbar^2 \omega} \rightarrow -\frac{im\sigma(\omega)}{\omega a \varepsilon_0} \approx \frac{e^2 \varepsilon_F m}{\pi (\hbar\omega)^2 a \varepsilon_0} = \frac{4\varepsilon_F (\hbar c)}{\pi (\hbar\omega)^2 a}$$

The dispersion relation is then given by:

$$\frac{\hbar\omega}{\varepsilon_F} = \left[\frac{800\alpha m}{a \varepsilon_F (\varepsilon_d + \varepsilon_m)} \right]^{1/2}$$

Note that the quasistatic plasmon relation for the cylinder (with only the intraband conductivity) may be written in terms of the same quantity for a sphere in the following way:

$$\hbar\omega_{m,\text{cylinder}} = \hbar\omega_{m,\text{sphere}} \times \left[\frac{m\varepsilon_m + (m+1)\varepsilon_d}{(m+1)(\varepsilon_d + \varepsilon_m)} \right]^{1/2}$$

Therefore, if $\varepsilon_m > 0$, the cylinder dispersion is always below the sphere dispersion. For $\varepsilon_m = \varepsilon_d = 1$, we get:

$$\hbar\omega_{m,\text{cylinder}} = \hbar\omega_{m,\text{sphere}} \times \left[\frac{2m+1}{2m+2} \right]^{1/2}$$

Dispersion for graphene coated system (intraband + interband)

If we introduce interband corrections, the dispersion will change to:

$$\frac{\varepsilon_d + \varepsilon_m}{m} = \frac{800\alpha}{\hbar\omega a} \left[\frac{\varepsilon_F}{\hbar\omega} + \frac{1}{4} \log \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega} \right]$$

Consistency with retarded result

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$$n_d J'_m(n_m ka) H_m^{(1)}(n_d ka) - n_m J_m(n_m ka) H_m^{(1)}(n_d ka)' - i \frac{\sigma(\omega)}{c\varepsilon_0} J'_m(n_m ka) H_m^{(1)}(n_d ka)' = 0$$

Therefore, by again only considering the Bessel function of the second kind in the Hankel function terms, we get:

$$n_d m (n_m ka)^{m-1} (n_d ka)^{-m} + m n_m (n_m ka)^m (n_d ka)^{-m-1} + i \frac{m^2 \sigma(\omega)}{c\varepsilon_0} (n_m ka)^{m-1} (n_d ka)^{-m-1} = 0$$

Which we readily simplify to give us:

$$n_d (n_m ka)^{-1} + n_m (n_d ka)^{-1} + i \frac{m \sigma(\omega)}{c\varepsilon_0} (n_m ka)^{-1} (n_d ka)^{-1} = 0 \rightarrow$$

$$\text{by multiplying the whole thing by } (n_m n_d ka) \rightarrow n_d^2 + n_m^2 + \frac{i m \sigma(\omega)}{\varepsilon_0 \omega a} = 0,$$

which is the non-retarded result derived before. Note that if we define m/a to be an effective momentum, q , we may recast the dispersion relation as follows:

$$\frac{\varepsilon_d + \varepsilon_m}{2} + \frac{iq\sigma(\omega)}{2\varepsilon_0\omega} = 0$$

Numerical comparison for graphene in the intraband approximation

To make a numerical comparison between the quasistatic and the fully retarded dispersions, we need an expression for the conductivity:

$$\frac{\sigma(\omega)}{c\varepsilon_0} = \frac{ie^2\varepsilon_F}{\pi\hbar^2c\varepsilon_0\omega} = 4i\alpha(\varepsilon_F/\hbar\omega)$$

Numerical comparison for graphene including interband contributions

If we introduce interband corrections, we have, instead,

$$\frac{\sigma(\omega)}{c\varepsilon_0} = 4i\alpha \left[(\varepsilon_F/\hbar\omega) + \frac{1}{4} \log \frac{2\varepsilon_F - \hbar\omega}{2\varepsilon_F + \hbar\omega} \right]$$

For $k \neq 0$

We start from the equation:

$$\frac{1}{\rho} \partial_\rho (\rho \partial_\rho \Psi(\mathbf{r})) + \frac{1}{\rho^2} \partial_\phi^2 \Psi(\mathbf{r}) + \partial_z^2 \Psi = 0 \rightarrow \rho \partial_\rho (\rho \partial_\rho \Psi(\mathbf{r})) - (\rho^2 k^2 + m^2) \Psi(\mathbf{r}) = 0$$

This is the modified Bessel equation. Therefore, we have that the general solution is given by (noting that the solution cannot be allowed to diverge at the origin):

$$\Psi(\mathbf{r}) = b_m I_m(k\rho) e^{im\phi + ikz} \text{ (for } \rho < a \text{) and}$$

$$\Psi(\mathbf{r}) = a_m K_m(k\rho) e^{im\phi + ikz} \text{ (for } \rho > a \text{)}$$

We have:

$$b_m I_m(ka) = a_m K_m(ka)$$

As well as:

$$-a_m \varepsilon_d K'_m(ka) + b_m \varepsilon_m I'_m(ka) = \frac{i\sigma(\omega)}{\omega \varepsilon_0 k} \left[\frac{m^2}{a^2} + k^2 \right] b_m I_m(ka)$$

Putting the two boundary conditions together gives us:

$$-\varepsilon_d I_m(ka) K'_m(ka) + \varepsilon_m I'_m(ka) K_m(ka) = -\frac{i\sigma(\omega)}{\omega \varepsilon_0 k} \left[\frac{m^2}{a^2} + k^2 \right] I_m(ka) K_m(ka)$$

We take, asymptotically, $I_m(x) \approx e^x / \sqrt{x}$, $K_m(x) \approx e^{-x} / \sqrt{x}$

The leading order, then, we get:

$$\varepsilon_d + \varepsilon_m + \frac{i\sigma(\omega)}{\omega \varepsilon_0 k} \left[(m/a)^2 + k^2 \right] = 0$$

$E = E_z$ **polarized plasmons**

For out of plane polarized plasmons, we have:

$$J_n(n_m ka) H_n^{(1)}(ka)' - n_m J'_n(n_m ka) H_n^{(1)}(ka) + i\sigma(\omega)/(c\varepsilon_0) J_n(n_m ka) H_n^{(1)}(ka) = 0$$

In the $ka \ll 1$ limit, this gives us the following condition:

$$-n(n_m ka)^n (ka)^{-n-1} - n n_m (n_m ka)^{n-1} (ka)^{-n} + i\sigma(\omega)/(c\varepsilon_0) (n_m ka)^n (ka)^{-n} = 0$$

From this, we obtain:

$$(ka)^{-1} + (ka)^{-1} = i\sigma(\omega)/(nc\varepsilon_0) \rightarrow 2 = \frac{ika\sigma(\omega)}{nc\varepsilon_0}$$

So we see that the imaginary part of the conductivity must be negative.

If we write the above condition in terms of the impedance, $Z_s \equiv -1/\sigma(\omega)$, we have:

$$\frac{1}{2} = -i \frac{nc\varepsilon_0}{ka\sigma(\omega)} \rightarrow Z_s = -\frac{i}{2} \frac{ka}{nc\varepsilon_0}$$